Week 4

- Bonus from last week
- Set Theory
- Kahoot

(b) (1 point) Compare the following two formulas. What are the differences?

$$P(x) \quad \text{and} \quad \forall x\, P(x).$$

Solution: First, in order for an interpretation to be suitable for the first formula, it needs to define x. For every interpretation for which the second formula is true, the first formula is also true, but the converse does not hold: Let us define the universe $U = \mathbb{N}$, $x = 2$ and the predicate $P(x) = 1 \iff x$ is even. It is easy to see that for this interpretation the first formula evaluates to true but not the second formula.

3. (2 points) $(\star\,\star)$

Find a universe U and define the predicates such that one of the following formulas is true but not the other. Try to come up with a very simple and compact solution, so you have enough time to explain and justify it.

(a)

$$\forall x\,(P(x) \to (\exists y \forall z\,(Q(z,y) \vee R(x,z))))$$

(b)

$$\exists y \forall x\,(P(x) \to ((\forall z\, Q(z,y) \vee (\forall z\, R(x,z))))).$$

Solution: Consider the universe $U = \mathbb{N}$. Furthermore, define

$$Q(x,y) = 1 \iff \text{“}x \text{ is even”}$$
$$R(x,y) = 1 \iff \text{“}y \text{ is odd”}$$
$$P(x) = 1.$$

$Q(x,y) = \begin{cases} 1 & , x \text{ even} \\ 0 & , x \text{ odd} \end{cases}$

With this choice the first formula is true but the second formula is not true.



Set Theory

$A = \{1, 2, 3\}$ $B = \{0, 1\}$

Cardinality: $|A| = 3$

Empty Set: $\emptyset = \{\}$ $|\emptyset| = 0$ $|\{\emptyset\}| = 1$

$\in$: $x \in A$: Predicate $\forall x\,(x \in A) \to \cdots$ predicate logic

$\forall x \in A$ ✓ complicated proofs

$A = B$: $\forall x\,(x \in A \leftrightarrow x \in B)$

$A \subseteq B$: subset $\forall x\,(x \in A \to x \in B)$ $\forall A\,(\emptyset \subseteq A)$

$A = B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$

$(A \subseteq B) \wedge (B \subseteq C) \Rightarrow A \subseteq C$

$A = \{1, 2, 3\}$ $B = \{0, 1\}$

$A \cup B$: union $\{x \mid x \in A \vee x \in B\}$ $\{0, 1, 2, 3\}$

$\mathcal{A} = \{\{1,2,3\}, \{2,4\}, \{0\}\}$

$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$ $\bigcup \mathcal{A} = \{x \mid x \in A \text{ for some } A \in \mathcal{A}\} = \{0, 1, 2, 3, 4\}$

$A \cap B$: intersection $\{x \mid x \in A \wedge x \in B\}$ $\{1\}$

$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$ $\bigcap \mathcal{A} = \{x \mid x \in A \text{ for all } A \in \mathcal{A}\}$

$B \setminus A$: $\{x \in B \mid x \notin A\}$ $\{0\}$

Theorem 3.4. *For any sets A, B, and C, the following laws hold:*

Idempotence:	$A \cap A = A$;
	$A \cup A = A$;
Commutativity:	$A \cap B = B \cap A$;
	$A \cup B = B \cup A$;
Associativity:	$A \cap (B \cap C) = (A \cap B) \cap C$;
	$A \cup (B \cup C) = (A \cup B) \cup C$;
Absorption:	$A \cap (A \cup B) = A$;
	$A \cup (A \cap B) = A$;
Distributivity:	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
Consistency:	$A \subseteq B \iff A \cap B = A \iff A \cup B = B$.

Power Set

$$\mathcal{P}(A) = \{S \mid S \subseteq A\}$$

$A = \{a, b\}$ $\quad \mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

$$|\mathcal{P}(A)| = 2^{|A|}$$

$A = \{a_1\ a_2 \cdots a_k\}$

$0 \quad 1 \cdots 1 \quad 2^k$

Cartesian Product

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

$$\emptyset \times A = \emptyset$$

$$\times_{i=1}^{k} A_i = \{(a_1, \ldots, a_k) \mid a_i \in A_i \text{ for } 1 \leq i \leq k\}$$

(a, b, x)

$A_1 \times A_2 \times A_3 = \times_{i=1}^{3} A_i \neq (A_1 \times A_2) \times A_3$ $\quad (a, b)$ $\quad ((a, b), x)$

$$|A \times B| = |A| \cdot |B|$$

Examples

$\emptyset \in \{\emptyset\}$ $\quad \emptyset \notin \{a, b\}$ $\quad \emptyset \subseteq$

- $\in$ vs. $\subseteq$

For each of the following choices of sets A and B, decide which of the statements $A \in B$ and $A \subseteq B$ are true.

i) $A = \{1, 0, \{0\}, 1\}, B = \{\{0, 1, 0, \{0, 0\}\}, 1, 10, 0\}$

ii) $A = \varnothing, B = \{\{\varnothing\}, \{\varnothing, \varnothing\}, \varnothing\}$

iii) $A = \{\{0\}, 0, \{0\}, \{\{0\}\}\}, B = \{0, \varnothing, \{\{0\}\}, \{0\}\}$

iv) $A = \{\varnothing\}, B = \{\varnothing, \{\varnothing, \varnothing, \varnothing\}\}$

i) $A \in B$ $\quad A \notin B$

ii) $A \in B$ $\quad A \subseteq B$

iii) $A \notin B$ $\quad A \subseteq B$

iv) $A \in B$ $\quad A \subseteq B$

Exam HS 20

1.) List the elements of the set $\{0, 1\} \times \{(0, 1)\}$. $\quad (0, (0,1)), (1, (0,1))$

2.) List the elements of the set $(\varnothing \cup \{\varnothing\}) \cap \{\varnothing, \{\varnothing\}\}$. $\quad \{\emptyset\} \cap \{\emptyset, \{\emptyset\}\} = \{\emptyset\}$ $\quad \{\emptyset\}$

3.) How many elements has the set $\mathcal{P}(\{\varnothing, \{\varnothing\}, \varnothing\} \cup \{\{\varnothing\}, \{\{\varnothing\}\}\})$? $\quad 8$

3 P / 115 P

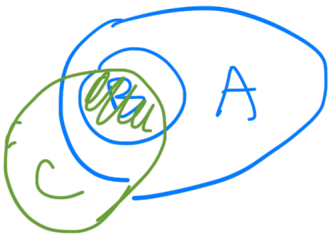
$\{\emptyset, \{\emptyset\}\}$

$\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}\}$

Prove or disprove:

If $B \subseteq A$ and $C \cap B \neq \emptyset$, then $C \subseteq A$

for any sets A, B, C.



$C = \{3, 4\}$

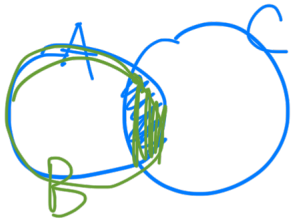
$B = \{4\}$

$A = \{4, 5\}$

Prove or disprove:

$(A \cup C = B \cup C) \wedge (A \cap C = B \cap C) \Rightarrow A = B$

for any sets A, B, C



Proof: We show $\forall x\,(x \in A \leftrightarrow x \in B)$

For any x,

$x \in A$

$\Leftrightarrow x \in A \wedge T$ $(F \wedge T \equiv F)$

$\Leftrightarrow x \in A \wedge (x \in C \vee x \notin C)$ $(F \vee \neg F \equiv T)$

$\Leftrightarrow (x \in A \wedge x \in C) \vee (x \in A \wedge x \notin C)$ (distr.)

$\Leftrightarrow (x \in A \cap C) \vee (x \in A \wedge x \notin C)$ (def. $\cap$)

$\Leftrightarrow (x \in A \cap C) \vee ((x \in A \wedge x \notin C) \vee \bot)$ $(F \vee \bot \equiv F)$

$\Leftrightarrow (x \in A \cap C) \vee ((x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin C))$ $(F \wedge \neg F \equiv \bot)$

$\Leftrightarrow (x \in A \cap C) \vee ((x \in A \vee x \in C) \wedge x \notin C)$ (distr.)

$\Leftrightarrow (x \in A \cap C) \vee ((x \in A \cup C) \wedge x \notin C)$ (def. $\cup$)

$\Leftrightarrow (x \in B \cap C) \vee ((x \in B \cup C) \wedge x \notin C)$ $(A \cap C = B \cap C \wedge A \cup C = B \cup C)$

$\Leftrightarrow \ldots\ldots$

$\Leftrightarrow x \in B$





